

ARTURO ESPINOSA VARGAS

Software Engineer | Google Cloud & Applied AI

Mexico City Metropolitan Area, Mexico | arturo.espinova@gmail.com

LinkedIn: [linkedin.com/in/arturo-espinosa-vargas-694a70239](https://www.linkedin.com/in/arturo-espinosa-vargas-694a70239)

GitHub: github.com/R2D2EV | Spanish: Native | English: Upper-Intermediate

PROFESSIONAL SUMMARY

Google Cloud Certified Associate Cloud Engineer with a background in software development, QA automation and technical teaching. Hands-on project experience deploying Python applications to Cloud Run and building retrieval-augmented generation (RAG) systems with FastAPI, PostgreSQL/pgvector and local LLMs. Combines a physics foundation with experience in automated testing, ETL pipelines and CI/CD to build maintainable cloud and AI applications.

Google Cloud Certified - Associate Cloud Engineer | August 2026

TECHNICAL SKILLS

Google Cloud - projects: Cloud Run, Cloud SQL for PostgreSQL, Cloud Build, Artifact Registry.

Google Cloud - hands-on labs: Compute Engine, GKE, VPC, IAM, Cloud Storage, load balancing, Cloud Logging, Cloud Monitoring; Terraform, Kubernetes.

AI & backend: Python, FastAPI, REST APIs, RAG, embeddings, semantic search, sentence-transformers, Ollama, Pydantic, SQLAlchemy; Azure OpenAI, Azure AI services.

Data & delivery: SQL, PostgreSQL, pgvector, JavaScript/Node.js, Docker, Docker Compose, Git, Jenkins, Linux, Bash, ETL, Power BI.

Quality engineering: pytest, Robot Framework, Selenium WebDriver, Cypress, Appium, Postman, CI/CD, TDD, BDD.

SELECTED GOOGLE CLOUD & AI PROJECTS

Simple Task App | Full-stack deployment on Google Cloud

Personal project | 2026 | FastAPI, PostgreSQL, Docker, Cloud Run, Cloud SQL

- Built a task application with an HTML/CSS/JavaScript frontend and FastAPI endpoints to create and list tasks, backed by persistent PostgreSQL storage.
- Containerized the application and deployed it to Cloud Run using Cloud Build and Artifact Registry; connected Cloud SQL for PostgreSQL through a Unix socket.
- Verified the public application and database persistence, exercising the complete workflow from local development to a managed cloud deployment.

GroundedDocs RAG | Document question answering with citations

Personal project | 2026 | Local implementation; ongoing development

- Implemented document ingestion, text extraction, chunking and semantic retrieval using FastAPI, sentence-transformers embeddings and PostgreSQL/pgvector.
- Integrated Ollama to generate answers from retrieved evidence, with source citations and abstention when documents do not provide sufficient support.
- Separated domain logic, use cases and infrastructure adapters; added document lifecycle operations, health/readiness checks, structured logging and pytest tests.

PROFESSIONAL EXPERIENCE

Solera | QA Automation & AI Engineer

2022 - August 2026

- Built and maintained automated test suites with Robot Framework, Selenium WebDriver, Appium and Cypress, reaching approximately 85% test coverage across supported applications.
- Developed JavaScript/Node.js ETL scripts for Jira and Tempo data and automated Excel/Power BI reporting, reducing report preparation from days to hours.
- Improved Jenkins CI/CD pipelines, increasing execution efficiency by approximately 20%.
- Built QA dashboards to track defects, team productivity and sprint health, informing release planning and resource allocation.
- Designed and deployed AI automation agents on AWS and Azure, including knowledge-base creation, model configuration and deployment.

UNAM, Faculty of Sciences | Teaching Assistant

2019 - Present | Ayudante de Profesor de Asignatura B

- Support Analytical Geometry courses for groups of 60+ students; develop exercises, guides and review sessions that explain mathematical concepts clearly.
- Redesign assignments to assess reasoning and conceptual understanding in the context of student use of generative AI.

Bootcamp Institute | Python Instructor

2025

- Designed and delivered project-based Python training in automation, testing and debugging; mentored teams building practical tools for business use cases.

ADDITIONAL AI PROJECTS

Audio-to-summary pipeline | Azure AI Foundry: Integrated Azure Speech-to-Text with Azure OpenAI GPT-4.1-mini to convert audio into text summaries.

Invoice extraction web application | Azure: Integrated the Azure Document Intelligence Python SDK with Flask and deployed the application to Azure App Service to extract structured invoice data.

Purchase prediction classifier | Azure ML: Trained and compared binary classifiers using Azure AutoML and modular workflows in Azure ML Designer.

CERTIFICATIONS

Google Cloud Certified: Associate Cloud Engineer | Issued August 17, 2026; expires August 17, 2029

Microsoft Certified: Azure AI Fundamentals | May 2026

Google Cloud: Cloud Computing Foundations Certificate | March 2026

Oracle Cloud Infrastructure: 2025 Certified Foundations Associate | November 2025; Certified AI Foundations Associate | September 2025

AWS Certified: Cloud Practitioner | March 2025

EDUCATION & PROFESSIONAL DEVELOPMENT

B.Sc. in Physics | UNAM: Mexico City, Mexico | 2020

In progress: Google Cloud Professional Cloud Architect preparation through Google Get Certified 3; Google TRI AI Program.